Geometry & Coordinate Geometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

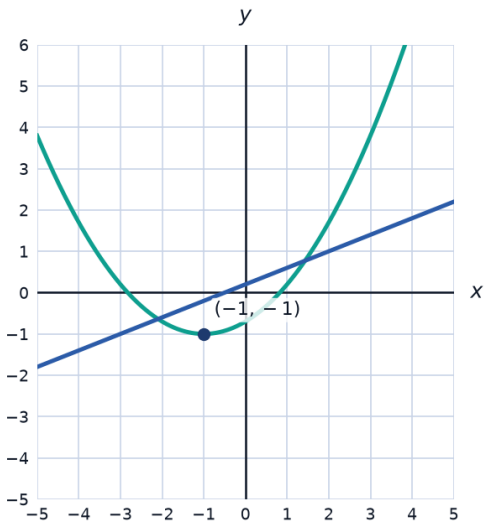
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Distance and midpoint**

Find the distance between $(1, 2)$ and $(4, 6)$.

2 **[Foundational]** • **Target: Distance and midpoint**

Find the midpoint of $(2, -3)$ and $(8, 5)$.



3 **[Foundational]** • **Target: Distance and midpoint**

Find the distance between $(-2, 1)$ and $(3, -11)$.

4 **[Foundational]** • **Target: Distance and midpoint**

Find the midpoint of $(-4, 7)$ and $(6, -3)$.

5 **[Foundational]** • **Target: Gradient (slope)**

Find the gradient of the line through $(1, 2)$ and $(5, 10)$.

6 **[Foundational]** • **Target: Gradient (slope)**

Find the gradient of the line through $(-3, 4)$ and $(2, -6)$.

7 **[Foundational]** • **Target: Gradient (slope)**

State whether the line through $(1, 1)$ and $(4, 1)$ is horizontal, vertical, or neither.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Equations of lines

Find the equation of the line with gradient 2 passing through $(3, 5)$, in the form $y = mx + c$.

9 [Intermediate] • Target: Equations of lines

Find the equation of the line through $(0, 4)$ and $(2, 10)$.

10 [Intermediate] • Target: Equations of lines

Find the x- and y-intercepts of the line $2x + 3y = 12$.

11 [Intermediate] • Target: Equations of lines

Write the equation of the line parallel to $y = 4x - 1$ passing through $(2, 9)$.

12 [Intermediate] • Target: Parallel and perpendicular lines

State the gradient of a line perpendicular to a line with gradient 5.

13 [Intermediate] • Target: Parallel and perpendicular lines

Determine whether the lines $y = 2x + 3$ and $y = 2x - 7$ are parallel, perpendicular, or neither.

14 [Intermediate] • Target: Parallel and perpendicular lines

Find the equation of the line perpendicular to $y = -\frac{1}{3}x + 2$, passing through $(3, 4)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The equation of a circle • Cross-Lesson: 3.01 Sequences & Series

Write the equation of a circle centered at the origin with radius 6.

16 [Challenging] • Target: The equation of a circle • Cross-Lesson: 4.01 Trigonometry

Write the equation of a circle centered at $(2, -3)$ with radius 5.

17 [Challenging] • Target: The equation of a circle • Cross-Lesson: 6.01 Limits & Derivatives

Find the radius and center of the circle $(x - 1)^2 + (y + 4)^2 = 49$.

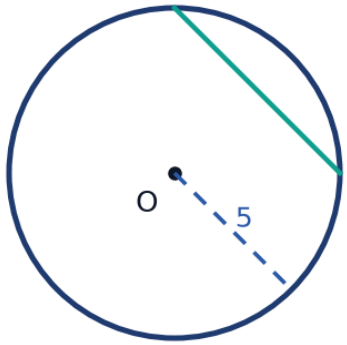
18 [Challenging] • Target: The equation of a circle • Cross-Lesson: 7.01 Integration & Applications

Find the equation of the circle centered at the origin that passes through $(3, 4)$.

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19 [Challenging] • Target: Circles: chords and arcs • Cross-Lesson: 8.01 Exponentials & Logarithms

Points A and B lie on a circle of radius 5 centered at O, with central angle $\angle AOB = 90^\circ$. Find the length of chord AB, using $AB = 2r\sin\left(\frac{\theta}{2}\right)$.



20 [Challenging] • Target: Circles: chords and arcs • Cross-Lesson: 9.01 Statistics & Probability

A chord of a circle with radius 10 cm is 12 cm long. Find the distance from the center to the chord.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Distance and midpoint

Find the distance between the two points shown.

a. $\sqrt{5}$

b. 4

c. 3

d. 5

22 [Foundational] • Target: Distance and midpoint

Find the midpoint of the segment joining the two points shown.

a. $(4, 3)$

b. $(2, 6)$

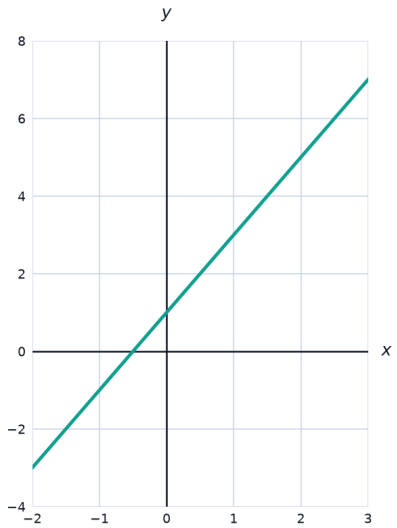
c. $(2, 3)$

d. $(8, 6)$

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23 [Foundational] • Target: Equations of lines

The graph shows a line. Find its equation in the form $y = mx + c$ by reading the graph.



a. $y = \frac{1}{2}x + 1$

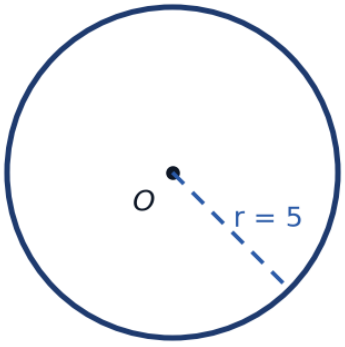
b. $y = 2x + 1$

c. $y = x + 1$

d. $y = 2x - 1$

24 [Foundational] • Target: The equation of a circle

Write the equation of the circle shown, centered at the origin.



a. $x^2 + y^2 = 10$

b. $x^2 + y^2 = 5$

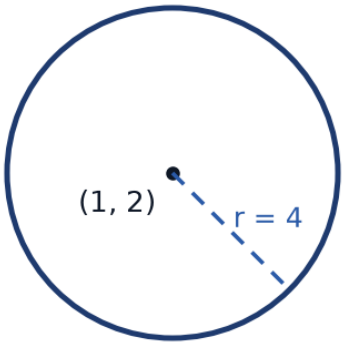
c. $x^2 + y^2 = 20$

d. $x^2 + y^2 = 25$

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25 [Foundational] • Target: The equation of a circle

The circle shown is centered at $(1, 2)$ with the radius marked. Write its equation.



a. $(x-1)^2+(y-2)^2=8$

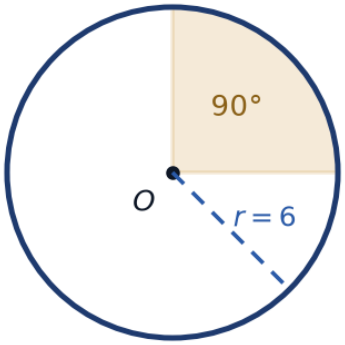
b. $(x+1)^2+(y+2)^2=16$

c. $(x-1)^2+(y-2)^2=16$

d. $(x-1)^2+(y-2)^2=4$

26 [Foundational] • Target: Circles: chords and arcs

Find the arc length of the 90° sector shown, in terms of π.



a. 12π

b. 3π

c. 1.5π

d. 6π

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27 [Foundational] • Target: Perimeter and area

A rectangle has length 15 and width 9, as shown. Find its perimeter and area.

a. $P = 135, A = 48$

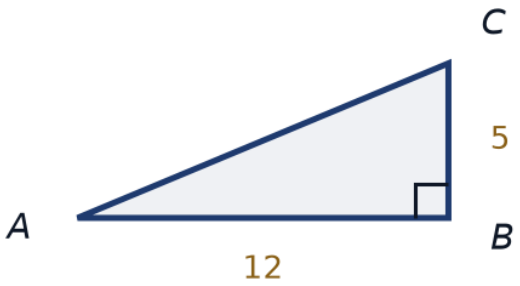
b. $P = 48, A = 135$

c. $P = 24, A = 135$

d. $P = 48, A = 24$

28 [Foundational] • Target: Perimeter and area

A right triangle has legs 5 and 12, as shown. Find its area and hypotenuse.



a. $A = 60$, hyp $= 13$

b. $A = 30$, hyp $= 17$

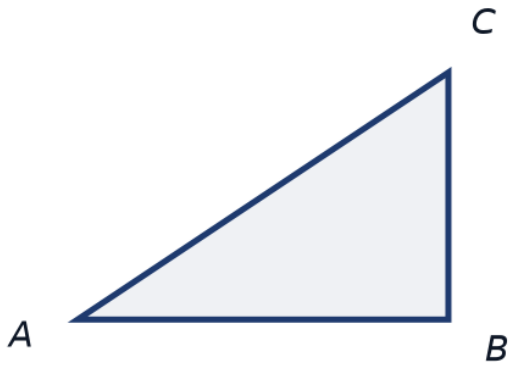
c. $A = 30$, hyp $= 13$

d. $A = 30$, hyp $= 12$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Coordinate geometry: identifying shapes

Is the triangle shown a right triangle?



a. Only if it is also isosceles

b. Cannot be determined

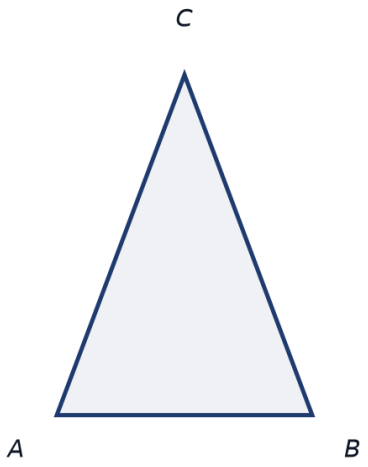
c. No

d. Yes

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30 [Intermediate] • Target: Coordinate geometry: identifying shapes

Is the triangle shown isosceles, scalene, or equilateral?



a. Isosceles

b. Cannot be determined

c. Equilateral

d. Scalene

31 [Intermediate] • Target: Parallel and perpendicular lines

Determine whether the lines $y = 2x + 1$ and $y = -0.5x + 4$ are parallel, perpendicular, or neither.

a. The same line

b. Parallel

c. Neither

d. Perpendicular

32 [Intermediate] • Target: The equation of a circle

Find the radius and center of the circle $(x + 2)^2 + (y - 5)^2 = 81$.

a. Center $(2, -5)$, radius 9

b. Center $(-2, 5)$, radius 9

c. Center $(-2, 5)$, radius 81

d. Center $(2, 5)$, radius 9

33 [Intermediate] • Target: Circles: chords and arcs

Points A and B lie on a circle of radius 8 centered at O, with central angle $\angle AOB = 60^\circ$. Find the arc length AB, in terms of π.

a. $\frac{16\pi}{3}$

b. $\frac{4\pi}{3}$

c. 16π

d. $\frac{8\pi}{3}$

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34 [Intermediate] • Target: Polygon angle sums

Find the sum of the interior angles of an octagon (8 sides).

a. 720°

b. $1{,}440^\circ$

c. 900°

d. $1{,}080^\circ$

35 [Intermediate] • Target: Polygon angle sums

Find the measure of each interior angle of a regular hexagon.

a. 140°

b. 135°

c. 108°

d. 120°

36 [Intermediate] • Target: Polygon angle sums

A polygon has an interior angle sum of $1{,}620^\circ$. Find the number of sides.

a. 10

b. 12

c. 9

d. 11

37 [Intermediate] • Target: Polygon angle sums

Find the sum of the interior angles of a decagon (10 sides).

a. $1{,}800^\circ$

b. $1{,}260^\circ$

c. $1{,}080^\circ$

d. $1{,}440^\circ$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Perimeter and area • Cross-Lesson: 7.01 Integration & Applications

A regular pentagon has side length 8, with area $A \approx 1.72s^2$. Find its area to the nearest whole number.

a. 110

b. 124

c. 96

d. 14

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39 [Challenging] • Target: Coordinate geometry: identifying shapes • Cross-Lesson: 8.01 Exponentials & Logarithms

Is the triangle with vertices $A(0, 0)$, $B(6, 0)$, $C(6, 4)$ a right triangle?

a. No

b. Cannot be determined

c. Yes

d. Only if it is also isosceles

40 [Challenging] • Target: Coordinate geometry: identifying shapes • Cross-Lesson: 9.01 Statistics & Probability

Determine whether the points $A(-3, 2)$, $B(1, 5)$, $C(4, 1)$ form a right triangle, using squared distances.

a. Only if it is also equilateral

b. No

c. Cannot be determined

d. Yes

41 [Challenging] • Target: The perpendicular bisector of a segment • Cross-Lesson: 1.01 Algebra & Equations

Find the equation of the perpendicular bisector of the segment joining $(-2, 1)$ and $(4, 9)$.

a. $y = -\frac{3}{4}x + \frac{23}{4}$

b. $y = -\frac{3}{4}x + 5$

c. $y = -\frac{4}{3}x + \frac{23}{4}$

d. $y = \frac{3}{4}x + \frac{23}{4}$

42 [Challenging] • Target: Word problems: geometry on a coordinate grid • Cross-Lesson: 2.01 Functions & Transformations

A city park is designed on a coordinate grid (in meters). Two gates are located at $(10, 20)$ and $(50, 60)$. Find the distance between the gates, to the nearest meter.

a. 113

b. 40

c. 57

d. 80

43 [Challenging] • Target: Word problems: geometry on a coordinate grid • Cross-Lesson: 3.01 Sequences & Series

A circular irrigation sprinkler is centered at $(8, 8)$ on a farm's coordinate grid (in meters) and has a radius of 15 meters. Write the equation of the circle it waters.

a. $(x + 8)^2 + (y + 8)^2 = 225$

b. $(x - 8)^2 + (y - 8)^2 = 15$

c. $(x - 8)^2 + (y - 8)^2 = 225$

d. $(x - 8)^2 + (y - 8)^2 = 30$

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44 [Challenging] • Target: Word problems: geometry on a coordinate grid • Cross-Lesson: 4.01 Trigonometry

A surveyor marks two points on a coordinate map (in km): a base camp at $(3, 4)$ and a summit at $(15, 4)$. Find the midpoint, to be used as a resupply station.

a. $(6, 4)$

b. $(9, 8)$

c. $(9, 4)$

d. $(18, 8)$

45 [Challenging] • Target: Geometry & Coordinate Geometry • Cross-Lesson: 6.01 Limits & Derivatives

Find the distance between $(3, 2)$ and $(-2, 14)$.

a. 15

b. 13

c. 17

d. 11

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